

ASHOKA UNIVERSITY

Department of Computer Science

FUEL UP+:
UNCERTAINTY-AWARE DEEP LEARNING FOR PERSONALIZED
GLUCOSE FORECASTING AND MEAL-SAFETY RANKING

An Undergraduate Thesis Report

submitted in partial fulfillment of the requirements for the degree of
B.Sc. (Hons.) in Computer Science and Entrepreneurship with Research

Submitted by

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Certificate by the Supervisor

This is to certify that the thesis report titled "Fuel Up+: Uncertainty-Aware Deep Learning for Personalized Glucose Forecasting and Meal-Safety Ranking" submitted by Arin Baswana (Student ID: 1020221410) is a bona fide record of work carried out under our supervision. To the best of our knowledge, this work has not been submitted elsewhere for the award of any degree or diploma and is fit to be considered for evaluation.

Prof. Lipika Dey

Date: _____

Prof. Partha Pratim Das

Date: _____

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Abstract

This report presents Fuel Up+, an extension of the earlier Fuel Up nutrition-tracking prototype into an uncertainty-aware system for short-horizon glucose forecasting and meal-safety ranking. The central claim is straightforward: for a system that intends to influence eating decisions, a point prediction alone is insufficient. A forecast that does not communicate uncertainty can appear precise while concealing clinically meaningful risk. To study this problem, three public continuous glucose monitoring datasets were harmonized into a common five-minute pipeline: HUPA-UCM for type 1 diabetes, BrisT1D for blood-glucose prediction in a competitive benchmark setting, and CGMacros for healthy, pre-diabetic, and type 2 diabetic participants with rich meal annotations. Across this unified corpus, GRU and Transformer models were trained in both absolute and delta-target modes and compared with Persistence, Linear Drift, Ridge Regression, and LightGBM baselines. Uncertainty was modeled in two ways: Monte Carlo Dropout and Pinball-loss Quantile Regression. The experiments show that model performance depends strongly on the data regime. GRU-delta performed better on the smaller type 1 diabetes datasets, while the Transformer became competitive on the nutritionally richer CGMacros dataset. More importantly, MC-Dropout produced narrow but poorly calibrated intervals, whereas Quantile Regression delivered materially better coverage at the cost of wider intervals. The report concludes that the practical value of Fuel Up+ lies not only in forecasting glucose, but in expressing uncertainty honestly enough to support safer meal comparisons through a prototype dashboard and safety-scoring interface.

1. Introduction

Fuel Up began as a conventional nutrition tracker. It logged meals, displayed calorie and macronutrient totals, and presented weekly trends. That version was useful as a record, but limited as a decision-support tool. The deeper question was never merely what a user had eaten, but what a particular meal was likely to do to that user's metabolic state in the near future. Fuel Up+ emerged from that shift in emphasis. The project moved from descriptive logging to predictive modelling.

This report studies whether a short-horizon glucose forecasting engine can be built from public continuous glucose monitoring data in a way that is methodologically disciplined, computationally feasible, and clinically honest about uncertainty. The work is interdisciplinary by design. It sits at the intersection of time-series modelling, physiological interpretation, and product design. The scientific challenge is to forecast glucose one hour ahead from six hours of prior glucose, meal, and activity history. The product challenge is to turn that forecast into a form that a user can act on without being misled by spurious precision.

The report is deliberately bounded. It does not claim a clinical device, a therapeutic recommendation engine, or a validated population-scale deployment. Its contribution is narrower and more defensible: a harmonized pipeline across three heterogeneous datasets, a comparative study of GRU and Transformer architectures, an explicit comparison of MC-Dropout and Quantile Regression for uncertainty estimation, and a prototype interface that converts model output into a meal-comparison workflow.

2. Background and Motivation

Blood glucose regulation is a dynamic physiological process shaped by carbohydrate absorption, gastric emptying, endogenous or exogenous insulin action, circadian variation, and physical activity. For non-diabetic individuals, the post-prandial rise is usually moderated by intact beta-cell function and returns toward baseline within a few hours. In diabetes, especially type 1 diabetes, the problem becomes more difficult because meal absorption and exogenous insulin uptake unfold on overlapping but variable timescales (Howey et al.; Ma et al.).

These dynamics explain why meal composition matters. Carbohydrate amount affects the magnitude of the rise, but fat, protein, fiber, and activity shape the timing and form of the excursion. A forecasting model that ignores those interactions may still fit average patterns, but it will miss precisely the cases that matter for decision support. That is the practical motivation for Fuel Up+: a user deciding between two meals needs a forecast that reflects both the expected response and the uncertainty around that response (Ma et al.).

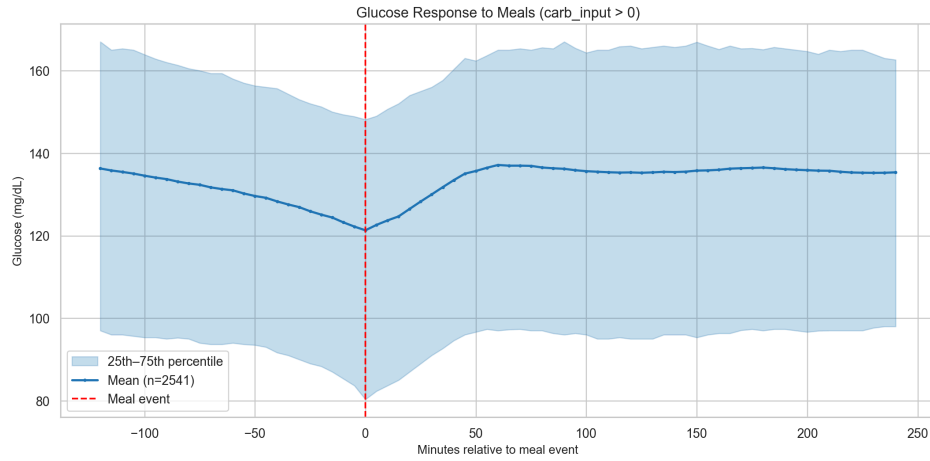


Figure 1. Meal-aligned glucose response in HUPA-UCM. The post-meal rise supports the choice of a 60-minute forecast horizon.

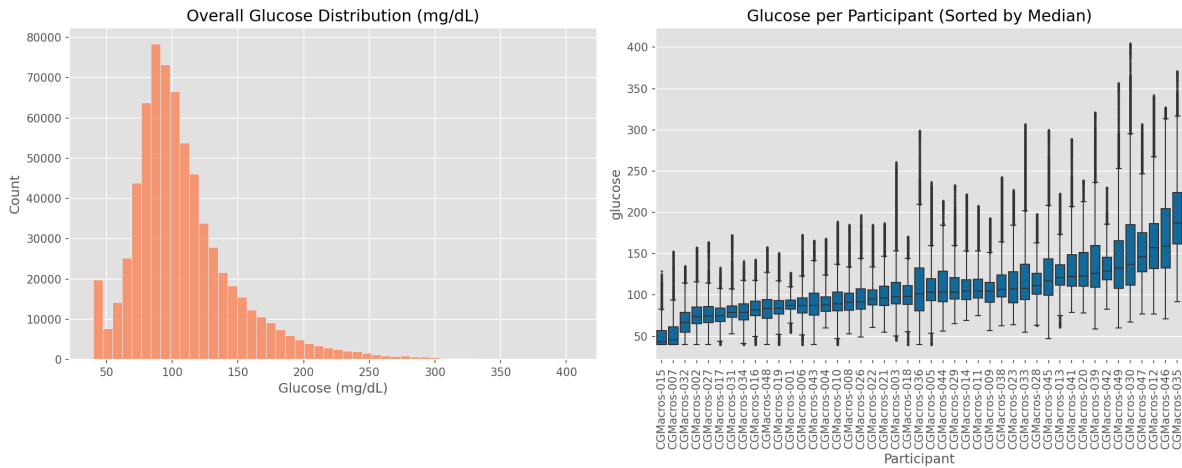


Figure 2. CGMacros glucose distributions. Participant-level spread highlights heterogeneity that a single global average cannot capture.

3. Literature Survey and Gap Analysis

The glucose-forecasting literature is rich in point-prediction studies and far thinner on uncertainty-aware evaluation. Recurrent models, especially GRUs and LSTMs, have remained strong baselines because public CGM datasets are usually small in participant count and recurrence imposes a useful bias toward temporal continuity. Transformers are attractive for multivariate time series because attention can weight distant parts of the history, but they are data-hungry and often unstable on narrow cohorts (Martinsson et al.; Cho et al.; Vaswani et al.).

A second strand of literature concerns uncertainty estimation. In clinical settings, calibration matters as much as average error. Deep ensembles, MC-Dropout, and Quantile Regression are all established approaches, but they serve different purposes. MC-Dropout mainly reflects model uncertainty. Quantile Regression is better suited to heteroscedastic data where the spread of the target varies from case to case. In free-living glucose data, that distinction is not academic. Meal timing, stress, sensor noise, and insulin kinetics make aleatoric uncertainty unavoidable (Lakshminarayanan et al.; Gal and Ghahramani).

Three gaps follow from this literature review. First, many studies are evaluated within a single dataset, so cross-cohort conclusions remain limited. Second, uncertainty is often reported in nominal rather than empirical terms, which hides miscalibration. Third, even when forecasting quality is good, the output is rarely translated into a user-facing decision interface. Fuel Up+ addresses all three gaps by harmonizing datasets, checking interval coverage directly, and connecting the forecast to a meal-comparison dashboard.

4. Problem Statement and Objectives

The problem addressed in this report is the following: given six hours of glucose, meal, and activity history, estimate the user's glucose one hour into the future and quantify the uncertainty of that estimate well enough to support a meal-safety ranking.

The objectives were to:

- harmonize HUPA-UCM, BrisT1D, and CGMacros into a common evaluation pipeline on a five-minute grid;
- compare GRU and Transformer models in both absolute and delta-target modes;
- benchmark deep models against simpler baselines to test whether added complexity was justified;
- compare MC-Dropout and Quantile Regression using empirical coverage rather than nominal claims alone;
- translate the resulting forecast into a prototype dashboard and safety score suitable for future user testing.

5. Scope, Methodology, and Design

The scope of the project is advisory forecasting. It does not include insulin dosing, closed-loop control, or clinical deployment claims. The modelling task was standardized across datasets by resampling to a five-minute grid and constructing 72-step windows representing six hours of

history with a target at $t + 60$ minutes. This horizon was chosen because it is both physiologically meaningful and practically relevant to meal choice.

Three feature scenarios were defined to test what information matters under different metabolic regimes. Scenario 1 used glucose history and meal carbohydrates, with insulin added for type 1 diabetes datasets. Scenario 2 extended the input to full macronutrients where available. Scenario 3 added activity signals such as heart rate, steps, and energy expenditure. This design allowed the experiments to test not only which model worked best, but also when extra nutritional or activity detail improved forecast quality.

Methodological discipline was especially important. Participant-level rather than row-level splits were enforced to prevent leakage from repeated windows belonging to the same individual. Scaling statistics were derived from training data only. Missingness was handled with short-gap forward filling, explicit mask channels, and rejection of windows spanning longer sensor gaps. These choices improve trustworthiness even when they reduce the final sample count.

At the model layer, the study compared GRU and Transformer encoders, each trained in absolute and delta modes. A delta target asks the model to predict the change in glucose rather than the future value directly. This is often a more informative learning target because it removes much of the trivial baseline signal and forces the model to represent dynamics.

6. Work Done: Implementation, Challenges, and Mitigations

The implementation was organized into three main code paths. The data and feature pipeline standardized heterogeneous raw inputs into a common tensor format. The model pipeline trained GRU and Transformer variants and evaluated them against baseline regressors. A third module added uncertainty estimation through MC-Dropout and Quantile Regression and exposed model outputs to a Flask-based web dashboard.

Several engineering challenges shaped the final design. The first was cross-dataset heterogeneity. HUPA and CGMacros supplied time-series data that had to be windowed from scratch, whereas BrisT1D arrived in a pre-windowed format and used mmol/L rather than mg/dL. The second challenge was missingness, which is unavoidable in free-living CGM data. The third was leakage control: in this domain, a careless row-level split can produce deceptively strong results by letting the model memorize individual baselines.

Hardware constraints also mattered. Training on full data with multiple seeds and wide hyperparameter sweeps was not practical within the time and compute budget of an undergraduate project. The study therefore used 10% subsamples for faster iteration, acknowledged the resulting performance ceiling, and treated comparative validity as more important than squeezing out the last unit of MAE. That decision was methodologically defensible because all models were evaluated under the same constraint.

The application layer translated the research output into a prototype. The dashboard displays the recent glucose history, the forecast trajectory, and an uncertainty band. A meal-comparison tool injects candidate meal profiles and ranks them using a safety score that penalizes both elevated predicted glucose and elevated upper-tail risk. This interface is not presented as a clinical

instrument. It is a demonstration of how uncertainty-aware forecasting can be communicated to a user.

7. Results and Discussion

The results show a coherent pattern. On the smaller type 1 diabetes datasets, GRU-delta was more reliable than the Transformer. On the larger and nutritionally richer CGMacros dataset, the gap narrowed and in some settings the Transformer became competitive or slightly better. This is not a contradiction. It reflects the fact that architecture performance depends on both data scale and feature richness.

Table 1. Representative test-set results from the report.

Dataset	Scenario	GRU-delta MAE	Transformer-delt a MAE	Unit
HUPA	S1	23.22	27.23	mg/dL
HUPA	S3	25.47	26.32	mg/dL
CGMacros	S2	10.92	10.58	mg/dL
CGMacros	S3	10.20	10.44	mg/dL
BrisT1D	S1	2.17	2.14	mmol/L
BrisT1D	S3	2.09	2.11	mmol/L

Two substantive conclusions follow. First, delta-target training was generally useful. It materially improved performance on CGMacros and remained at least competitive elsewhere. Second, richer inputs helped only where physiology suggested they should. Full macro and activity information improved performance on CGMacros, where endogenous insulin and meal composition drive much of the variation. The same additions did not improve HUPA to the same extent, which is biologically plausible because type 1 diabetes outcomes are dominated by exogenous insulin kinetics.

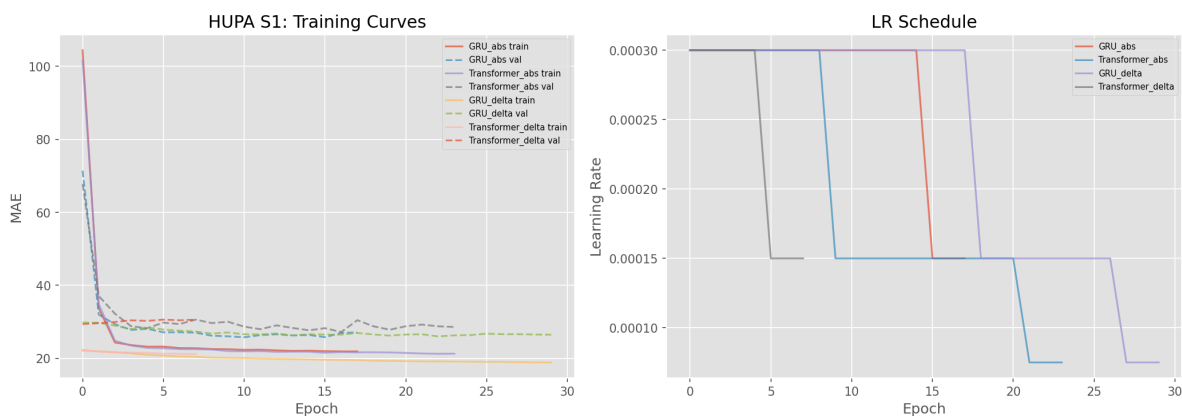


Figure 3. Training curves on HUPA S1. The Transformer begins to overfit earlier than the GRU, consistent with the smaller cohort size.

Uncertainty estimation was the most important qualitative result. MC-Dropout produced comparatively sharp intervals but missed the true value too often. On HUPA, observed coverage was 44.6% against an 80% nominal target. On BrisT1D S3, the gap was even larger: 22.2% observed coverage for MC-Dropout versus 64.1% for Quantile Regression. Quantile Regression did not achieve perfect calibration either, but it was substantially closer to the target and therefore more defensible for a user-facing system.

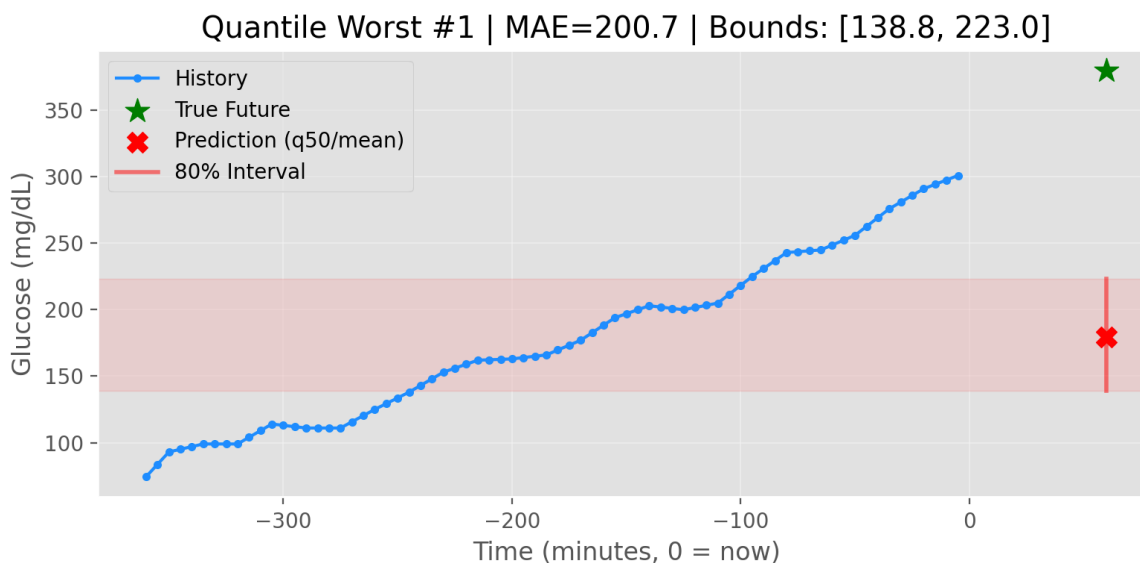


Figure 4. Quantile-based uncertainty on a difficult HUPA case. The interval is wide, but it better reflects risk than a narrow and overconfident bound.

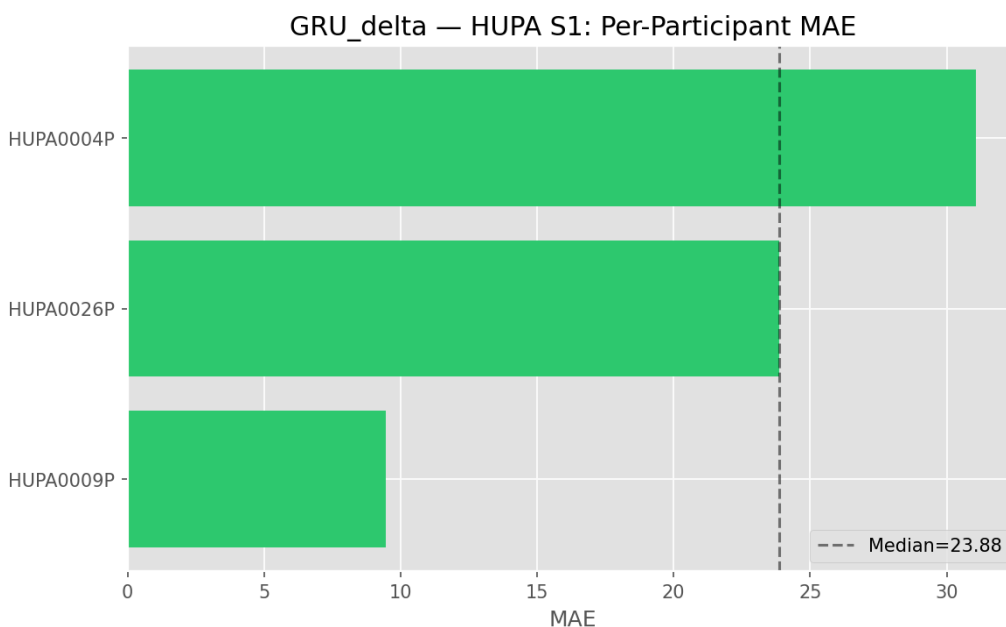


Figure 5. Per-participant MAE on HUPA S1. The spread across participants suggests that future personalization should include user-specific adaptation.

The report also retains three important caveats. The experiments were run on 10% subsamples to make iteration feasible. Most deep runs used a single seed. And the public datasets are largely European and American, so transferability to South Asian dietary patterns remains untested. These limitations do not invalidate the comparative findings, but they should temper any broader deployment claim.

8. Conclusions

Fuel Up+ demonstrates that a student nutrition tracker can be turned into a serious forecasting prototype when the problem is reframed around uncertainty rather than around logging alone. The strongest empirical conclusion is that interval quality matters as much as point accuracy. A model that is slightly less sharp but materially more honest is the safer foundation for a decision-support system. Within the current data regime, GRU-delta is the stronger default choice for smaller type 1 diabetes datasets, while Transformer models become more attractive as sample richness increases.

The broader contribution of the work is methodological. The project establishes a reproducible way to compare architectures, targets, and uncertainty heads across heterogeneous public CGM datasets while keeping leakage controls explicit. That is a meaningful outcome for an undergraduate thesis because it produces a usable research scaffold, not just a single set of benchmark numbers.

9. Extensions and Future Work

The next stage of this work should prioritize methodological consolidation before product expansion. The most immediate step is full-data retraining with multiple random seeds so that the performance claims rest on tighter estimates. A second priority is post-hoc calibration or conformalization of the quantile head to improve finite-sample coverage. A third is lightweight personalization, such as on-device fine-tuning or adaptation after a user has contributed a small amount of personal CGM history.

A later stage could examine an Ashoka-specific pilot, but only under clear ethical and institutional review. Such a pilot would be useful primarily as a data-collection and behavior-study exercise, not as a clinical deployment. Beyond that, any Indian scaling claim requires new local data. The present report gives reasons to believe personalized glucose forecasting is promising, but it does not justify national generalization from non-Indian cohorts.

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Publications and Communications

No peer-reviewed publication has resulted from this work at the time of submission.